

Pluto Nano (HamGeek, AliExpress): random reboots and how to fix them

Zynq-7010 + AD9363 board without SD slot · Vendor firmware v0.39-dirty, U-Boot v0.20 (Oct 9 2025) · Fix: Tezuka v0.3.22 firmware for Nano

Symptoms

- While streaming, the Pluto randomly drops off the PC: sometimes after one minute, sometimes after ten.
- The serial console shows a **kernel panic** followed by a full reboot (U-Boot appears).
- It happens with both usb: and ip:, with any software, cable or USB port.

Diagnosis

- **Serial console** (Nano UART connector, /dev/ttyUSB0 at 115200 baud): several kernel panics were captured, each in a different kernel function (timers, USB FunctionFS, activity LED), always with corrupted pointers. This is the pattern of **random memory corruption**, not a specific software bug.
- **Pluto load and memory**: CPU almost idle, free memory stable. Overload ruled out.
- **Idle RAM test** (repeated md5sum of a 150 MB file in /tmp): no errors. The failure only shows while streaming, with DMA, USB and CPU all busy.
- **Identical case in the Tezuka project** (issue #383 in F50E0/tezuka_fw): a Nano with the same vendor U-Boot had the same panics. Changing only the kernel did not help; **changing the bootloader (boot.frm) did**.

Cause

The vendor bootloader (FSBL/U-Boot, partition qspi - fsbl-uboot) configures the Zynq DDR memory controller. Everything points to that configuration being marginal: it works most of the time, but occasionally corrupts memory under real load.

Fix

Install the **Tezuka firmware for Nano**, which ships its own bootloader (different RAM configuration and U-Boot 2025.01), kernel 6.12 and FPGA logic. Result: hours of streaming without a single crash.

Steps

Before you start

- The Nano **has no SD slot**, so the internal flash must be written. Rewriting the bootloader is risky; if it failed, only JTAG could recover the board.
- **Do not use Tezuka v0.3.21**: it has a kernel corruption bug fixed in v0.3.22.
- You need sshpass, dfu-util and picocom. Keep the serial console open during the whole process:

```
picocom -b 115200 --logfile ~/pluto_uart.txt /dev/ttyUSB0
```

1. Full flash backup

With the Pluto running normally. Both checksums on each line must match.

```
P="sshpass -p analog ssh -o StrictHostKeyChecking=no \  
  -o UserKnownHostsFile=/dev/null root@192.168.2.1"  
mkdir ~/pluto_backup && cd ~/pluto_backup  
for i in 0 1 2 3; do $P "cat /dev/mtd$i" > mtd$i.bin; done  
for i in 0 1 2 3; do  
  echo "mtd$i $(md5sum < mtd$i.bin | cut -c1-32) $($P "md5sum /dev/mtd$i" | cut -c1-32)"  
done
```

2. Download and verify the firmware

The dd extracts the U-Boot environment embedded in boot.frm. Check that the md5 sums match the ones shown.

```
wget https://github.com/F50E0/tezuka_fw/releases/download/v0.3.22/\
tezuka-nano-v0.3.22-d99264f.zip
echo "ee4440cf3b32a476783b930e0747ca8cc15b2e6372e746ad2931aabeeb6360af \
tezuka-nano-v0.3.22-d99264f.zip" | sha256sum -c -
unzip tezuka-nano-v0.3.22-d99264f.zip
cd tezuka-nano-v0.3.22-d99264f/flash
dd if=boot.frm of=u-boot-env.bin bs=1 skip=752976 count=131072
md5sum boot.dfu u-boot-env.bin
# ab90d363807aaff1836888c76a430c28 boot.dfu
# 827b7b18129745dd5a3031a6efc82ad7 u-boot-env.bin
```

3. Enter DFU mode

You should see 0456:b674 with five partitions: boot.dfu, firmware.dfu, u-boot-extra-env.dfu, u-boot-env.dfu and spare.dfu.

```
$P 'device_reboot sf'
lsusb | grep 0456          # 0456:b674 = DFU
sudo dfu-util -l
```

4. Write the three pieces, in this order

Each command must end with Done! (the "Invalid DFU suffix" warning on the third one is expected).

```
sudo dfu-util -a firmware.dfu -D pluto.dfu
sudo dfu-util -a boot.dfu -D boot.dfu
sudo dfu-util -a u-boot-env.dfu -D u-boot-env.bin
sudo dfu-util -a u-boot-env.dfu -e
```

- **All three are required.** The vendor U-Boot cannot decompress Tezuka's LZMA kernel (Unimplemented compression type 3): flashing only pluto.dfu will not boot.
- **Finish with u-boot-env.dfu.** If the last write is firmware.dfu, the old U-Boot saves its environment on exit and overwrites the new one.

5. Boot

Unplug and replug. The console should show U-Boot 2025.01, Linux version 6.12.77, AD936x ... successfully initialized and Welcome to Tezuka. The jffs2 warnings on the first boot are normal: Tezuka reformats the old settings partition. The root password is still analog.

Recovery

As long as U-Boot works you can always go back: enter DFU from the serial console (press a key during boot and type run dfu_sf) and write the mtd*.bin backups back to their partitions.